

CS 736: Medical Image Computing

(Instructor : Suyash P. Awate)

Mid-Semester Examination. Closed Book/Notes.

Duration: 2 hours

Roll Number: _____

Name: _____

When in doubt, make suitable assumptions, state them clearly, and proceed to solve the problem.
All the best!

Formulae for Reference

Poisson probability mass function: $P(x; \lambda) := \lambda^x \exp(-\lambda)/x!$ with mean = λ , variance = λ , and mode = $\text{floor}(\lambda)$.

Rician probability density function: $P(x; \nu, \sigma) := \frac{x}{\sigma^2} \exp(-\frac{x^2 + \nu^2}{2\sigma^2}) I_0(\frac{x\nu}{\sigma^2}), \forall x \geq 0$

Gamma probability mass function: $P(x; \alpha, \beta) := \eta(\alpha, \beta) x^{\alpha-1} \exp(-\beta x)$, where $\eta(\alpha, \beta)$ is a normalization constant that is a function of parameters α and β . In this case, mean = α/β , variance = α/β^2 , and mode = $(\alpha - 1)/\beta$.

Rayleigh probability density function: $P(x; \sigma) := \frac{x}{\sigma^2} \exp(-\frac{x^2}{2\sigma^2}), \forall x \geq 0$

1. [30 points] Consider a standard K-means problem with N data points $\{x_i \in \mathbb{R}^D\}_{i=1}^N$ in a D -dimensional Euclidean space to be clustered into K groups. Let the set of means across all groups be $\{\mu_k \in \mathbb{R}^D\}_{k=1}^K$.

Now we introduce constraints, one for each of the k means, such that the k -th mean μ_k must lie within a hyperplane given by the equation $a_k^\top \mu_k = 0$. The parameter $a_k \in \mathbb{R}^D$ for each hyperplane is specified.

Your goal is to find the clustering/grouping and the set of means that satisfies all the constraints.

- (20 points: 5 + 5 + 5 + 5) Solve the underlying constrained optimization problem for the clustering (i.e., for the constrained means and the clustering/grouping both) by using the method of Lagrange multipliers to handle the constraints. Specify the Lagrangian objective function. Mathematically derive the updates for all parameters/variables being optimized over; write clear pseudo code for an implementable algorithm.

- (10 points: 5 + 5) Solve the same problem **without** using the method of Lagrange multipliers. Specify the objective function (with a clear mathematical derivation/justification). Mathematically derive the updates for all parameters/variables being optimized over; write clear pseudo code for an implementable algorithm.

2. [30 points] Consider the problem of edge-preserving denoising of a 2-dimensional image where the noise model is known to be independent and identically distributed (i.i.d.) Gaussian (zero mean, unit variance) and the Markov random field (MRF) prior relies on 4-neighborhoods along with the Huber function.
- (5 points) Specify the Huber function mathematically. Specify its derivative function mathematically.

- (25 points: 5 + 5 + 5 + 5 + 5) Write clear pseudo-code for performing iterative image denoising with the goal of seeking the maximum-a-posteriori estimate of the noiseless image.

Write pseudo-code for functions to compute the log-likelihood and its derivative, for the entire image.

Write pseudo-code for functions to compute the log-prior (upto some additive constant) and its derivative, for the entire image.

Write pseudo-code for iterative parallel updates inspired by gradient descent (as discussed in class), including the code for ensuring that any update never worsens the objective-function value; this pseudo-code should call the functions written earlier.

Additionally, if the Huber function was replaced by a squared-error function, then write pseudo-code for functions to compute the log-prior (upto some additive constant) and its derivative, for the entire image.

3. [10 points: 5 + 5] Explain graphically (by drawing a picture) the minorization maximization scheme underlying the expectation maximization (EM) algorithm, i.e., within each iteration of the EM algorithm, specify the design philosophy of the $F(\cdot)$ function in relationship with the log-likelihood function.

Mathematically justify that maximizing the $F(\cdot)$ function typically improves (and can never worsen) the parameter estimation.

